

Fraction division (1)

Unit three · fraction $\div$ integer + fraction $\div$ fraction · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5B Unit 3 + Modern Education 5L B Unit 11

Core traps: ☐ T3 Fraction division - forgetting to invert the divisor after the division sign ($\div$ becomes $\times$ reciprocal)
• T9 The result is not reduced

SSPA association: ● High frequency The sub-test must be taken every year, accounting for about 12-18% of Paper 1

Prerequisite knowledge: Course 7 (common denominators with different denominators) · Course 9 (multiplication of fractions, including mixed numbers $\times$ fractions)

Goals of this course: ❶ Fraction $\div$ whole number ($= \times$ reciprocal) ❷ Fraction $\div$ fraction (reverse after division sign) ❸ Division of mixed numbers ❹ Division word problems

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 4 question , 5 minutes)

#	Question	Difficulty	Working Space (Show full working)
1	Write the reciprocal of $\frac{2}{3}$. The countdown.	Basic	
2	Write what are the reciprocals of $\frac{1}{5}, \frac{3}{4}, \frac{5}{2}$ What are the reciprocals of ?	Basic	
3	Convert $2\frac{1}{3}$ into an improper fraction.	Basic	
4	Calculation: $\frac{3}{4} \times \frac{1}{2} = ?$ and reduce the results. (Review multiplication of fractions)	Basic	

II.Core Knowledge + Worked Examples

Knowledge Point 1: Fraction $\div$ Integer ● SSPA

① Core rule: fraction $\div$ integer = fraction $\times$ (reciprocal of integer)

② For example: $\frac{2}{3} \div 2 = \frac{2}{3} \times \frac{1}{2} = \frac{2}{6} = \frac{1}{3}$

③ Reciprocal of integer $a = \frac{1}{a}$ (remember: $a = \frac{a}{1}$, the reciprocal is $\frac{1}{a}$) ④ After division, it must be reduced! The numerator and denominator are divided by the greatest common factor (HCF) at the same time) $\frac{1}{a}$

④ It must be reduced after division! Divide both the numerator and denominator by the highest common factor (HCF)

Fraction $\div$ integer = fraction $\times$ reciprocal of integer

$\frac{2}{3}$

$\frac{1}{3}$

$\div 2 = \times \frac{1}{2}$ (reciprocal)

$\frac{2}{3} \times \frac{1}{2} = \frac{2}{6}$

$= \frac{1}{3} \checkmark$

Half $\rightarrow$ 2 parts and only 1 part left = $\frac{1}{3}$

Example 1

calculate: $\frac{3}{4} \div 2 = ?$

Example 2

calculate: $\frac{5}{6} \div 4 = ?$

Knowledge Point 1 Synchronization practice (write out: ① turn $\times$ reciprocal ② numerator $\times$ numerator $\cdot$ denominator $\times$ denominator ③ simplify)

#	Question	Difficulty	Working Space
5	$\frac{2}{3} \div 2 = ?$		
6	$\frac{5}{8} \div 5 = ?$		
7	$\frac{4}{7} \div 3 = ?$		
8	$\frac{9}{10} \div 6 = ?$		

⚠ The most frequent error: $\div 2$ is treated as $\times 2$ instead of $\times \frac{1}{2}$! Remember: $\div \text{Integer} = \times \text{The reciprocal of that integer}$.

Knowledge Point 2: fraction $\div$ fraction = $\times$ reciprocal ● SSPA Must-Test

Core formula: $\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{a \times d}{b \times c}$

- ① The divisor after the division sign **must be inverted (the numerator and denominator are swapped)** ② The division sign becomes a multiplication sign ③ Then calculate by fraction multiplication ④ Must reduce
Don't make a mistake: only the fraction after the division sign is inverted!
⑤ **Make no mistake: only the fraction after the division sign is inverted!**

The fraction after the division sign needs to be "reversed" - only the divisor is reversed!

$$\frac{2}{3} \div \frac{1}{4} \overset{\text{turnover}}{\times} \frac{4}{1} = \frac{8}{3} = 2\frac{2}{3}$$

Rule: $a/b \div c/d = a/b \times d/c$

Only the fraction after $\div$ is reversed!

❑ Trap detonation example 3 (the most important demonstration in this class)

calculate: $\frac{2}{3} \div \frac{1}{4} = ?$

✗ Common mistakes (60% students)

$$\frac{2}{3} \times \frac{1}{4} = \frac{2}{12} = \frac{1}{6}$$

Forgot to reverse the divisor of 14! Treat it directly as a multiplication sign.

✓ Correct solution

$$\frac{2}{3} \times \frac{4}{1} = \frac{8}{3} = 2\frac{2}{3}$$

$\div 14 \rightarrow \times 41$ (the fraction after the division sign must be reversed!)

💡 Tip: "Invert the division sign and turn it into a multiplication sign. The numerator is multiplied by the numerator and the denominator is multiplied by the denominator. The answer must be simplified!"

Example 4

calculate: $\frac{3}{5} \div \frac{2}{3} = ?$

Knowledge Point 2 Synchronization practice (must write out: ① $\div \rightarrow \times$ ② divide number inversion ③ simplify)

#	Question	Difficulty	Working Space
9	$\frac{1}{2} \div \frac{1}{3} = ?$		
10	$\frac{3}{4} \div \frac{1}{2} = ?$		
11	$\frac{5}{6} \div \frac{2}{3} = ?$		
12	$\frac{4}{5} \div \frac{3}{10} = ?$		

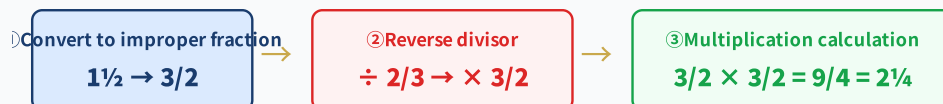
 **High-frequency trap:** Just remember "reverse" but also reverse the previous scores! For example, $23 \div 45$ is incorrectly written as 32×54 — wrong! Only what follows $\div$ is inverted.

Knowledge Point 3: Division with Mixed Fractions SSPA Advanced

Standard three-step method:

- ① **mixed fraction $\rightarrow$ improper fraction**(integer $\times$ denominator + numerator)
- ② **Reverse the divisor after the division sign**, the division sign becomes the multiplication sign
- ③ **Multiply the numerator by the numerator and the denominator by the denominator** $\rightarrow$ Reduction $\rightarrow$ Convert the improper fraction to a mixed fraction

Division of mixed numbers: first convert to improper fraction $\rightarrow$ then reverse $\rightarrow$ then multiply



①

Convert Improper Fractions

Convert all mixed numbers into improper fractions

②

divisor reversal

$\div$ The following fraction is reversed

③

Multiplication calculation

Numerator $\times$ Numerator
Denominator $\times$ Denominator

④

Reduction • Return

Reduction + Improper Fraction $\rightarrow$ Mixed Fraction

Example 5

Calculate: $1\frac{1}{2} \div \frac{2}{3} = ?$

Example 6

Calculate: $2\frac{1}{3} \div 1\frac{1}{2} = ?$

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
13	$1\frac{1}{4} \div \frac{1}{2} = ?$		
14	$2\frac{1}{2} \div 1\frac{1}{3} = ?$		
15	$1\frac{2}{3} \div \frac{3}{4} = ?$		

✗ Trap: Mixed fractions are reversed before they are false.

$$1\frac{1}{2} \div \frac{2}{3} = \frac{1}{2} \times \frac{3}{2} \quad \text{✗}$$

Just separate the numerator and denominator of $1\frac{1}{2}$ – big mistake!

✓ Correct: Convert improper fractions first

$$\frac{3}{2} \times \frac{3}{2} = \frac{9}{4} = 2\frac{1}{4}$$

$$1\frac{1}{2} = \frac{3}{2} \rightarrow \frac{3}{2} \div \frac{2}{3} = \frac{3}{2} \times \frac{3}{2} = \frac{9}{4}$$

Knowledge Point 4: Fraction Division Application Questions ● SSPA Required Exam

Four-step method to solve the problem: ① **Circle keywords**("Share equally" "Each..." "How many servings can..." = division)

② **Column formula**(Total amount $\div$ per serving = number of servings; Total amount $\div$ number of servings = per serving)

③ **Calculate according to the division steps** ④ **Write a complete answer**(Step points will be deducted if you do not write an answer!)

Key word question: Total quantity $\div$ Per portion = Number of divisible portions

Total quantity $\frac{5}{6}$

Each section $\frac{1}{4}$ m

$\frac{1}{4}$

= ? Sections

$$\begin{aligned} &\frac{5}{6} \div \frac{1}{4} \\ &= \frac{5}{6} \times \frac{4}{1} \\ &= 3\frac{1}{3} \text{ sections } \checkmark \end{aligned}$$

Example 7

$\frac{3}{4}$ A cake is divided equally among 5 people. What fraction of the cake does each person get?

Example 8

A ribbon is $\frac{5}{6}$ meters long, cut into sections every $\frac{1}{4}$ meters. How many pieces can it be cut into?

Knowledge Point 4: Synchronized practice (column → calculate → answer sentence)

#	Question	Difficulty	Working Space
16	$\frac{1}{2}$ A pancake is divided equally among 3 people. What fraction of the pancake does each person get?		
17	A bottle of juice contains $\frac{4}{5}$ liters. Pour one cup for every $\frac{2}{5}$ liter. How many cups can be filled?		
18	A rope is $2\frac{1}{2}$ meters long and cut into sections every $\frac{1}{2}$ meter. How many pieces can it be cut into?		

III. Lesson Layered Synchronization Practice

Basic layer (total 4 questions, everyone must do)

#	Question	Difficulty	Working Space
19	$\frac{3}{5} \div 3 = ?$		
20	$\frac{4}{7} \div 2 = ?$		
21	$\frac{3}{8} \div \frac{1}{4} = ?$		
22	$\frac{5}{6} \div \frac{5}{12} = ?$		

Advanced layer (total 3 questions,   choose do)

#	Question	Difficulty	Working Space
23	$\frac{7}{9} \div \frac{2}{3} = ?$		
24	$1\frac{1}{5} \div \frac{3}{5} = ?$		
25	$\frac{9}{10} \div \frac{3}{5} = ?$		

#	Question	Difficulty	Working Space
26	$\frac{3}{4} \div \frac{2}{3} \div \frac{1}{2} = ?$ (Continuous division of three fractions - calculate the first two first, then $\div$ the third result)		
27	$2\frac{1}{4} \div 1\frac{1}{2} = ?$		

IV. Ultimate challenge area

#	Question	Difficulty	Working Space
 1	A rope is $3\frac{1}{2}$ meters long. Use half of it first, and then cut the remaining $\frac{1}{4}$ meters into a section. How many pieces can it be cut into? (You need to find the remaining length first, and then $\div$ the length of each paragraph)		
 2	A box of candy weighs $\frac{3}{4}$ kilograms, and each $\frac{1}{8}$ kilogram contains a small package. After filling 3 bags, how many bags of candy can be left?		

#	Question	Difficulty	Working Space
A1	$\frac{2}{3}$ Kilograms of sugar, one pack per $\frac{1}{9}$ kg. How many bags can be packed?		
A2	The original water bottle contains $\frac{3}{4}$ liters. After drinking $\frac{1}{3}$ liters, pour the remaining water into 2 cups. How many liters of water are in each cup?		

V. Class afterhomework

Basic must doquestion (total 6 questions, mustwrite out complete step:
 $\div \rightarrow \times \rightarrow$ reverse $\rightarrow$ simplify)

#	Question	Difficulty	Working Space
H1	$\frac{2}{5} \div 2 = ?$		
H2	$\frac{3}{8} \div 3 = ?$		

H3	$\frac{1}{2} \div \frac{1}{4} = ?$		
H4	$\frac{3}{4} \div \frac{1}{3} = ?$		
H5	$\frac{7}{10} \div \frac{2}{5} = ?$		
H6	$1\frac{1}{2} \div \frac{1}{2} = ?$		

For advanced, choose doquestion (total 2 questions,  choose do)

#	Question	Difficulty	Working Space
H7	$2\frac{1}{3} \div 1\frac{1}{4} = ?$		
H8	There are $\frac{3}{5}$ liters in a bottle of orange juice. Pour a small glass for every $\frac{3}{20}$ liter. After filling 2 cups, how many more cups can be filled?		

VI. The Lessoncorecommon errorssummary

☒ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	$\div$ integer is treated as $\times$ integer : $\frac{3}{4} \div 2 = \frac{6}{4}$	$\div 2 = \times \frac{1}{2}$, not $\times 2$! The reciprocal of an integer is $1/\text{integer}$
2	$\div$ Fraction forgetting to reverse the divisor : $\frac{2}{3} \div \frac{1}{4} = \frac{2}{12}$	$\div \frac{1}{4} = \times \frac{4}{1}$, only the one after the division sign is inverted!
3	Reverse the previous fraction as well : $\frac{2}{3}$ $\div \frac{4}{5} = \frac{3}{2} \times \frac{5}{4}$	Only fractions after $\div$ are inverted! The front is unchanged, the back is reversed.
4	The answer is not reduced : $\frac{6}{12}$ as the final answer	Must be reduced to the simplest fraction (numerator and denominator $\div$ HCF)
5	Improper fractions are not converted to mixed fractions	Numerator $>$ Denominator $\rightarrow$ Convert into mixed numbers (subject to the requirements of the division test)

6	Mixed fractions are divided directly by $\div$ without conversion: $1\frac{1}{2} \div 2$	You must first convert to an improper fraction ($1\frac{1}{2} = \frac{3}{2}$), and then follow the division steps
7	Application question "number of copies vs. each copy" confusion	Total amount $\div$ amount per serving = number of servings; total amount $\div$ number of servings = amount per serving. Distinguish clearly!

Lam Fung Academy • LF Academy • We don't teach math. We teach trap avoidance.

 Related topics: L09 Multiplication of fractions • L23 Mixing of three fractions • Related topics: L02 Two-digit division • L03 Three-digit division • L05 Four word problems